

Python qDMR codeplug generator

Motivation and Approach

All codeplug generation programs (CPS) known to me run on Windows, and creating a custom codeplug is a laborious process. As a Linux user, I sought alternatives and discovered [qdmr by DM3MAT](#). *qdmr* can utilize the Repeatermap, Repeaterbook, and Radiod APIs to populate repeater entries. Since March 2026, the first two API providers have required an individual access token to prevent bulk downloads. Even so, repeaters must still be entered individually and assigned to zones. After some research, I found the program [dmrfill](#), which addresses this issue but—unfortunately—has not yet implemented the necessary access tokens for API queries. The developer is aware of the [problem](#), but it remains unresolved.

I adopted the core concept of *dmrfill*, expanded upon it, and implemented it in Python. Since *qdmr* employs device-specific extensions, it is necessary to copy these from an existing codeplug and use them as a template. At the top level, *qdmr* uses the following objects: 'version', 'settings', 'radioIDs', 'contacts', 'groupLists', 'channels', 'zones', 'roamingChannels', 'roamingZones', 'positioning', 'commercial', and 'sms'. With the exception of 'channels' and 'zones', all elements are retained unchanged; the channels and zones are then regenerated and inserted. Dependencies may exist between these objects; consequently, they must either be removed from the template codeplug, or the target object of a dependency must not be deleted. This applies to 'settings' (which might contain zone and channel defaults) and 'positioning' (APRS settings that reference channels). The deletion of APRS channels is automatically prevented.

Installation and Codeplug Preparation

A current Python 3 interpreter and the following libraries are required: copy, csv, functools, itertools, json, math, operator, Path, pickle, requests, sys, time, and yaml. It is advisable to set up a virtual environment and install the missing libraries there.

```
Create virtual environment
python3 -m venv venv

# Activate environment
source venv/bin/activate

# Install packages
python3 -m pip install -r requirements.txt
```

The "src" directory should also be copied to the current directory. The file "pydmrfill.py" should be made executable. Afterward, you can exit the virtual environment if desired.

```
# Deactivate when done
deactivate
```

Next, the codeplug should be read from the target device using qdmr. Preferred settings can be configured on the device itself or within qdmr beforehand; however, dependencies on specific channels and zones should be avoided. The codeplug from the device may also be empty. It is advisable to create two template channels—one for FM and one for DMR. These channels must begin with the string "Template". If these templates are missing, the program's built-in static templates will be used instead. These templates allow for the configuration of talkgroups, channel bandwidths, squelch settings, transmit power, APRS beacons, etc. The name and location of this template codeplug are passed to the program as parameters (see below).

Usage

First, activate the virtual environment (see above), switch to the "src" directory, and run pydmrfill.py:

```
./pydmrfill.py [config.yaml]
```

If the first parameter (configuration file) is omitted, the filename "PyDMRFill.yaml" is used.

The output is written to the file "out.qdmr.yaml" in the local directory and can also be opened from the command line using qdmr:

```
qdmr out.qdmr.yaml
```

Parameters

The parameter file uses the YAML format and should not contain tabs; indentation is achieved using spaces.

General

These parameters apply globally.

User-Agent	Default for the Repeaterbook API; a custom User-Agent can be defined but should include an email address.
API-Token	Assigned by Repeaterbook upon request.
FM-Bandwidth	Default value used if the repeater's bandwidth is undefined. <i>Wide</i> or <i>Narrow</i> .
TS1-TA	Talker Alias for Timeslot 1 (OpenGD77). <i>Off</i> , <i>APRS</i> , <i>Text</i> , or <i>APRS+Text</i> .
TS2-TA	Talker Alias for Timeslot 2.
CP-Template	Template file for the codeplug.
KEEP-ZONES	List of zone names that must not be deleted from the template file.

Countries (Zones)

Countries are requested from the Repeaterbook API via separate queries and cached to avoid hitting API query limits. The final zone name is generated by combining the country name and the region name.

Mode	Controls the caching of API queries. <i>Default</i> queries the API without caching; <i>Load</i> uses the cache and avoids the API query; <i>Dump</i> queries the API and also permanently saves the results.
Country	Country name used for the API query.
Regions	List of regions that must be located within the specified country.

Regions

Regions are circles around a specific location with a defined maximum distance (radius). The label can be chosen freely but should be kept short so that the resulting zone names on the device do not become excessively long.

Latitude	Geographic latitude in degrees (floating-point number).
Longitude	Geographic longitude in degrees (floating-point number).
MaxDistance	Radius in km.

For each combination of zone and region, two zones are created on the device—one containing FM channels and the other DMR channels. Within these zones, the channels are arranged according to their distance from the region's center point, in ascending order.

Additional Channels (CSV Import)

To include additional channels not listed on Repeatermap, channel lists can be imported as CSV files. This includes direct frequencies and hotspots. Each CSV file is assigned a separate zone.

Sample Configuration (Example)

```
User-Agent: channel creator 0.1, <callsign@darc.de>
API-Token: app_XXXXXXX
FM-Bandwidth: Wide           # Default FM-Bandwidth if not defined (Small or Wide)
TS1-TA: APRS                 # Timeslot 1 DMR Talker Alias (Off, APRS, Text, APRS+Text)
TS2-TA: APRS                 # Timeslot 2 DMR Talker Alias (Off, APRS, Text, APRS+Text)
CP-Template: dt878uv2-sample.yaml # Codeplug-Template file
KEEP-ZONES: [Seefunk, C23-direkt]
Zones:
  DL:
    Mode: Load               # Should be 'Default', 'Load' or 'Dump'
    Country: Germany          # Country name is used for Repeatermap query, must match there
    Regions:                  # List of regions in given country
    VR:
      Latitude: 54.2038585    # Center of repeater search circle, lat and lon
```

```
Longitude: 12.8183909
MaxDistance: 100      # Radius of search circle in kilometers
LL:
  Latitude: 48.0606517
  Longitude: 11.0412388
  MaxDistance: 150
SP:
  Mode: Load
  Country: Poland
  Regions:
    Swinemuende:
      Latitude: 53.9109417
      Longitude: 14.2694135
      MaxDistance: 100
OZ:
  Mode: Load
  Country: Denmark
  Regions:
    Klintholm:
      Latitude: 54.9524922
      Longitude: 12.4666164
      MaxDistance: 100
SM:
  Mode: Load
  Country: Sweden
  Regions:
    Trelleborg:
      Latitude: 55.4377801
      Longitude: 13.2224797
      MaxDistance: 100
OE:
  Mode: Load
  Country: Austria
  Regions:
    Innsbruck:
      Latitude: 47.2713648
      Longitude: 11.1999975
      MaxDistance: 100
CSV-Import:
  Free:
    Path: CSV-free/Channels.csv
  Simplex:
    Path: CSV-Simplex/Channels.csv
  Hotspot:
    Path: Hotspot/Channels.csv
```